# AI-Powered Data Collection & Analytics Platform — PRD

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# Product Requirements Document

## AI-Powered Data Collection & Analytics Platform

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## 1. Overview

### 1.1 Summary

A web-based platform where people can submit information (through reports, forms, or automated feeds), have that information checked for accuracy and completeness, and then get it turned into clear summaries,



trends, and dashboards using AI. The platform is both a place to collect data and a place to make sense of it — instead of raw records piling up unread, the system reads them for you and tells you what they mean.

### 1.2 Problem Statement

Teams collect information — surveys, feedback, operational reports, structured submissions — but struggle to turn it into decisions quickly. Reviewing everything by hand is slow, inconsistent, and doesn't scale as volume grows. Existing dashboards show numbers but rarely *explain* them in plain language, and most intake forms don't check submissions carefully enough to keep the resulting data trustworthy.

### 1.3 Goals

- Provide a reliable way to collect information — through forms, uploaded reports, and automated feeds — with rigorous checking before it's accepted.
- Guarantee the data is accurate and complete before it's stored or analyzed.
- Generate accurate, easy-to-understand AI summaries and analytics from the collected information. • Give Analysts and Admins a dashboard to explore trends, dig into individual records, and (for Admins) export findings.
- Build a system that is secure, auditable, and cost-aware around its use of AI.

### 1.4 Non-Goals

- Not a general-purpose business intelligence tool where anyone can freely query any data however they like — the platform is purpose-built around structured intake and AI-generated insight. • Not a real-time streaming system — near-real-time updates are sufficient for v1. • Not building custom AI models from scratch — v1 uses established, hosted AI services.

## 2. Target Users & Personas

| Persona | Description | Core Needs |
|---|---|---|
| **Submitter:** | Person or system filing a report or submitting data via a form, upload, or automated feed | Fast, clear, low-friction submission with immediate feedback if something's wrong; ability to ask about the status of their own submissions in plain language |
| **Analyst** | Views the dashboard and works with data **after** it has been reviewed and refined by AI; cannot access raw/pending submissions or export data | |
| **Admin** | Everything an Analyst can do, plus managing users, configuration, and data export | |

Trustworthy, AI-refined summaries; filterable dashboards with charts and graphs; the ability to dig into the records behind any insight; a way to ask ad-hoc questions instead of building a report from scratch Ability to add/manage users, configure how data is collected and validated, see how the AI is being used and what it costs, and export data that Analysts cannot

## 3. User Stories

**Report/Data Submitter**

- As a **Submitter**, I want to feed data into the platform by uploading a report (file) or filling out a direct-entry form, so that I have a channel that fits how my data already exists.
- As a **Submitter**, I want immediate feedback as I fill out a form, so I know right away if something is missing or wrong, rather than finding out after I submit.
- As a **Submitter**, I want to upload a report or file and see clear, specific errors for anything that doesn't check out, before it's accepted.
- As a **Submitter**, I want to ask "what's the status of my submission?" or "why was my upload rejected?" and get a direct answer scoped to my own data, without needing analyst-level dashboard access.

**Dashboard User (Analyst & Admin)**

- As a **user viewing the dashboard**, I want to see charts and line graphs summarizing the data, so that I can spot trends and patterns visually instead of reading raw records.
- As an **Analyst or Admin**, I want to type a question about the data ("what changed this month?", "which area had the most rejected submissions?") into a chat box and get a clear, written answer with links back to the records or data behind it.

**Analyst**

- As an **Analyst**, I want to interact only with data that has already been reviewed and refined by AI, so that the information I base decisions on has already been checked, cleaned, and summarized — not raw, unprocessed submissions.
  - **–** *Submissions are not visible to Analysts until AI review is complete; pending or raw records are out of reach regardless of how the dashboard is filtered.*
- As an **Analyst**, I want an AI-generated weekly summary of new (refined) submissions highlighting notable trends or outliers.

**Admin**

- As an **Admin**, I want to provision and manage user accounts, so that I can control who has access to the platform and what role they hold.
- As an **Admin**, I want to do everything an Analyst can do — including exporting data — so that I retain full analytical and reporting capability myself, on top of my administrative duties. **–** *Export is an Admin-exclusive capability; Analysts can view, drill down, and query, but cannot export.*
- As an **Admin**, I want to define what a valid submission looks like (which fields, what's required, what the rules are) without needing a developer's help.
- As an **Admin**, I want to see which AI summaries were generated, from what data, and at what cost. • As an **Admin**, I want to ask things like "how many submissions failed review this week and why?" and get a plain-language answer instead of building a report.

## 4. Functional Requirements

**4.1 Data Input**

- **Multiple ways to submit:** dynamic forms, report/file upload, and automated feeds from other systems.
- **Report ingestion:** submitters can feed data in via a formal report upload, not just simple forms — it's checked against the same rules a form submission would be, so reports and forms end up producing

the same trustworthy, consistent data.



• **Configurable forms:** Admins define what fields exist, what's required, and what the rules are; forms update automatically to match — no need to wait on a development cycle for a new field. • **Draft/autosave:** longer forms save progress automatically so nothing is lost partway through. • **Batch upload:** uploaded files are checked row by row before anything is accepted, with a clear, downloadable report of anything that didn't pass.

### 4.2 Validation (Data Checking)

- **Checked on entry and again on the way in:** submitters get immediate feedback as they type or upload, and everything is re-checked when it reaches the system — so a person can't accidentally bypass the checks.

• **Rule-based checks:** required fields, valid ranges, correct formats, and allowed values are all enforced. • **Cross-field and business-rule checks:** e.g., "field B is required if field A is X," duplicate detection, and consistency checks against existing records.

- **Protection against bad or malicious input:** anything submitted is cleaned and checked to prevent harmful or corrupted data from getting in.
- **Clear error messages:** submitters see specifically what's wrong and where, not a generic "submission failed."
- **Nothing is silently dropped:** anything that fails a check is quarantined with a clear reason, and is visible to Admins for review and correction rather than disappearing.

### 4.3 Data Storage

- Submitted information is stored in an organized, structured way that can flex as new types of fields or reports are added, without needing everything rebuilt from scratch each time.

• A full history is kept of who submitted what, when, and any changes made afterward. • Old data can be retired or removed according to a defined retention policy (useful for compliance requirements).

### 4.4 AI Analysis Pipeline

- **When analysis runs:** on a regular schedule (e.g., nightly or weekly), on demand (an Analyst or Admin asks for a fresh summary), and incrementally as new data comes in.
- **How a summary gets made:**
    1. **Preparation** — new submissions are cleaned up, deduplicated, and organized into a form that's ready to analyze.
    2. **Number-crunching** — trends, changes, and outliers are calculated directly from the data, as hard facts, before the AI ever writes anything.
    3. **AI narration** — the AI is given those calculated facts (plus a representative sample of records) and writes a clear, human-readable summary, highlighting key findings and anything unusual. 4. **Sourcing** — every claim in an AI summary links back to the specific records or figures that produced it, so it can be independently checked rather than taken on faith.
    5. **Packaging** — the output is organized into summary cards, trend callouts, and confidence indi cators, alongside the full written narrative.
- **Consistency:** every summary records exactly what version of the AI and instructions were used to produce it, so results are traceable and comparable over time.
- **Human feedback:** Analysts can flag a summary as inaccurate, and that feedback is used to improve future summaries.
- **Analyst access gate:** a submission is not visible or usable by Analysts until it has completed AI review. Until then, it's in a "pending review" state — visible to the Submitter (as a status only, e.g. "processing") and to Admins (who can inspect the raw record if needed for troubleshooting), but not to Analysts. This applies everywhere — dashboards, exports, and chat questions alike; there is no way for an Analyst to reach unreviewed data.

### 4.5 Conversational Query Interface (Ask-a-Question Chat)

A chat-style “ask a question, get an answer” feature, available to everyone but scoped to what each person is allowed to see. This is a core feature for launch, not a nice-to-have — it’s the fastest way to get from “I have a question” to “I have an answer.”

- **Who can ask what:**
  - **Submitters** can ask about their own submissions only (“what’s the status of my last upload?”, “why was this rejected?”) — never anyone else’s data.
  - **Analysts** can ask broader analytical questions (“what changed in the West region this month?”, “show me submissions similar to last week’s anomalies”).
  - **Admins** can additionally ask operational questions (“how many users signed up today?”, “what’s our AI spend this month vs. last?”).
- **Always grounded in real data:** a question is never answered by having the AI guess or make things up. Instead, the question is turned into a real, permission-scoped lookup against the actual data; the AI is only ever shown the real result and asked to explain it in plain language — it cannot see or reach data outside what that person is allowed to access.
- **Honest when it doesn’t know:** if a question can’t be safely or reliably answered, the system says so plainly rather than guessing.
- **Remembers context:** follow-up questions in the same conversation build on what came before, the same way a conversation with a person would.
- **Where it lives:** a persistent chat panel on the dashboard (for Analysts/Admins) and on a Submitter’s own status page; answers include text and, where it helps, a small chart.
- **Fast:** answers to common questions typically arrive in under about 5 seconds.

### 4.6 Analytics & Dashboard

- **Overview dashboard:** submission volume over time, pass/fail rates, and top categories at a glance.
- **Drill-down:** click any insight, chart, or data point to see the specific records behind it. • **Charts and graphs as a core part of the dashboard, not an afterthought:**
  - **Line/trend charts** — how submission volume, pass/fail rates, and AI usage change over time, with adjustable date ranges.
  - **Bar charts** — comparisons across categories or regions (e.g., submissions by area, rejection reasons by frequency).
  - **Distribution charts** — for numeric data, showing spread and outliers rather than just averages.
  - **Breakdown charts** — status splits (accepted/rejected/pending) or category share of the total. **– Highlighted anomalies** — anything the AI flags as unusual is called out directly on the relevant chart, linking back to the explanation behind it.
- **Interactive:** hover for exact values, click to drill into records, zoom into a time range. • **Consistent look and feel:** charts across the dashboard follow the same visual language, so results are easy to compare and read at a glance.
- **Works well on any screen size**, including smaller devices.
- **Chart-aware chat answers:** when a question naturally calls for a visual (“show me volume by week”), the chat answer includes a small chart alongside the written explanation, matching the look of the main dashboard.
- **Export:** charts and reports can be exported as images or documents — **export is Admin-only**; Analysts can view, filter, and drill down, but cannot export.
- Saved views and filters, available to both Analyst and Admin.
- Configurable alerts (e.g., “notify me if the rejection rate exceeds 10%” or “notify me when the AI flags something unusual”).

### 4.7 Admin & Configuration

• **User management:** Admins add, edit, deactivate, and assign roles to user accounts. • **Full Analyst capability, plus more:** everything an Analyst can do on the dashboard, an Admin can also do — Admin isn't a separate, cut-off role, it's a superset of Analyst.



- **Export (Admin-exclusive):** Admins can export data and reports; this is intentionally withheld from the Analyst role.
- **Configuration:** defining what fields exist, what's required, and what validation rules apply — with a history of changes over time.
- **Usage visibility:** a view into how much the AI is being used and what it's costing, broken down over time.

**Role permission summary:**

| Capability | Submitter | Analyst | Admin |
|---|---|---|---|
| Submit via form/report/feed | Yes | – | – |
| View own submission status | Yes | – | – |
| View AI-refined data & dashboard (scoped to what they can see) | – | Yes | Yes |
| View raw/pending (unrefined) submissions | – | Yes | Yes |
| Ask questions via chat (own data) | – | Yes | Yes |
| Drill down into records | – | Yes | Yes |
| Export data/reports/charts | – | – | Yes |
| Manage users | – | – | Yes |
| Configure data rules | – | – | Yes |

## 5. Non-Functional Requirements

What "quality" means for this platform, in plain terms:

| Category | What it means in practice |
|---|---|
| **Speed** | Forms respond instantly as people type; the system stays responsive even during a large batch upload |
| **Reliability at scale** | The platform can handle bursts of activity (e.g., a flood of submissions) without slowing down or dropping data |
| **Availability** | The system should be up and accepting submissions essentially all the time — losing incoming data is treated as far more serious than a slow dashboard |
| **Security** | Information is protected both while it's moving through the system and while it's stored; access follows a "only what's needed" principle |
| **Privacy & compliance** | Sensitive fields are identifiable and handled with appropriate care; data can be retained or removed according to policy; every action is logged for accountability |
| **Accessibility** | The platform is usable by people relying on assistive technology, following recognized accessibility standards |

| Category | What it means in practice |
|---|---|
| **Transparency into operations** | If something goes wrong anywhere in the pipeline, it's traceable — from submission, through checking, through AI review |
| **Cost discipline** | AI usage is monitored and capped so costs stay predictable rather than growing unchecked |

## 6. AI Summary Reliability & Guardrails

This deserves real product attention, not just an engineering afterthought — trust in the AI's output is the whole value proposition:

1. **Facts first, narration second** — the AI explains numbers that were already calculated; it never invents or "does the math" itself.
2. **Everything is sourced** — every insight links back to the records or figures that produced it, so it can be independently checked.
3. **Confidence is signaled** — the system flags when a sample is small or the data is noisy, rather than presenting every summary with equal confidence.
4. **People can correct it** — Analysts can flag an inaccurate summary, and that feedback improves future results.
5. **Everything is traceable** — a given summary can always be traced back to exactly what version of the AI and instructions produced it.
6. **Cost and speed are managed** — usage is capped and reused where possible, and spend is visible to Admins rather than being a surprise.

## 7. Success Metrics

| Metric | Target |
|---|---|
| Errors caught before submission vs. after | > 90% caught immediately, before the person even submits |
| Rejected/quarantined submission rate | < 2% of submissions after the first 60 days |
| Time from data arriving to a summary being available | Under 5 minutes for incremental updates, under 30 minutes for scheduled rollups |
| AI summary accuracy (how often flagged as wrong) | < 5% of summaries flagged as inaccurate |
| Chat answer accuracy (how often flagged as wrong) | < 5% of chat answers flagged as inaccurate |
| Chat response time | Under 5 seconds for common questions |
| Weekly active Analysts/Admins on the dashboard | Tracked post-launch to measure adoption |
| Share of questions resolved via chat vs. manual digging | Tracked as a signal the feature is actually saving people time |
| AI cost per 1,000 submissions analyzed | Defined and tracked from week one, with a budget ceiling |

## 8. Milestones / Rollout Plan

Phase Scope

**Phase 0 — Foundations** User accounts and roles, basic configuration tools, groundwork for everything else
**Phase 1 — Intake & Validation** Forms, checks on entry and on arrival, file upload with row-level checking, quarantine flow for
anything rejected
**Phase 2 — Storage & Audit** Full history/audit trail, versioned configuration, feed-based submission
**Phase 3 — AI Pipeline v1** Fact-calculation layer, AI-written summaries (scheduled and on-demand), sourcing/citations
**Phase 4 — Dashboard & Analytics** Overview dashboard, drill-down, exports, alerts **Phase 5 — Ask-a-Question Chat** Chat panel for Submitters/Analysts/Admins; questions scoped to what each person can see;
conversation memory; same sourcing as summaries
**Phase 6 — Hardening** Cost controls, accuracy testing (including chat), accessibility review, load testing, faster incremental
updates

## 9. Risks & Open Questions

• **AI getting numbers wrong** — mitigated by always having the AI narrate pre-calculated facts rather than compute them itself, but this needs to be tested rigorously before launch, not after. • **Rules changing over time** — how are older submissions handled when the rules for what's "valid" change? Needs a clear policy, not case-by-case handling.
• **Cost predictability** — AI usage costs can scale unpredictably with volume; needs budget caps and alerts from day one.
• **Handling of sensitive/personal information** — depends heavily on what kind of data is being collected; needs a classification pass before the intake rules are finalized.
• **Which AI provider to use** — needs a decision based on reliability, cost, and capability relative to expected volume.
• **Single organization vs. multiple organizations** — does this platform serve one organization, or many separate ones sharing the same system? This materially affects how data is kept separate and isn't fully decided yet.
• **Making sure chat questions can't "leak" data** — the chat feature must never let a clever question reveal data the asker isn't supposed to see. This needs to be explicitly tested, not just assumed to work.

## 10. Open Decisions Needed From Stakeholders

1. Expected volume of submissions at launch and over the first year — this drives sizing and cost planning. 2. Will this serve a single organization, or multiple separate organizations on one platform? 3. Any compliance requirements (e.g., data protection regulations) that affect how long data is kept, how it's protected, and what needs to be logged.

4. Any preference for which AI provider to use, if there’s an existing relationship or requirement. 8